



Factors Associated with Thoughts of Migration Among Latinx Mental Health Professionals from Puerto Rico to Mainland United States

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Abstract

Highly skilled professionals in low-income countries often migrate to other countries to seek improvements in their living conditions, incomes, and quality of life. This rapidly growing phenomenon, also known as the “brain drain,” refers to the migration of workers to countries with more opportunities. This study examines various “push” and “pull” factors for the growing migration of Puerto Rican healthcare professionals to the mainland United States. Using a quantitative and cross-sectional approach, we collected data on mental health professionals’ socioeconomic and work-related needs. We conducted a binary logistic regression to identify whether socioeconomic (i.e., age, marital status, gender, highest level of education, and income satisfaction) and work-related factors (i.e., perceived organizational support and emotional exhaustion related to the workplace) contributed to having thoughts of migration to the United States in the year prior completing the survey. Almost half of the participants (48.9%) reported having thoughts of migration. Factors associated with migration thoughts included younger age, dissatisfaction with current income, emotional exhaustion in the workplace, and lower perceived organizational support. The migration of healthcare workers is a major aspect of globalization, shaping the circumstances in which some countries gain and others lose opportunities for social and economic development. Recommendations to retain this highly needed group of professionals in their places of origin and support them in their potential migration to the U.S. mainland are highlighted.

Keywords Brain drain · Latin America · Puerto Rico · Mental health care professionals · Migration · United States

The study of social and economic conditions leading to the overall migration of individuals has become an increasingly important area of public health research and practice (Castañeda et al., 2015). One area of particular concern is the need to understand how socioeconomic and work-related factors are contributing to a growing socio-economic phenomenon, known as the “brain drain” of healthcare professionals. This refers to the migration of highly trained

workers from resource-limited areas, where poverty related stressors are prevalent and pervasive to destinations with more opportunities (Botezat & Ramos, 2020; Zubaran, 2012). Although the migration of healthcare workers is one of the most widely studied issues in healthcare worldwide (Bludau, 2021), existing literature have focused on the migration of medical doctors (Ighobor, 2016), and nurses (Peters et al., 2020), with less attention placed on migration patterns of mental health professionals.

The link between precarious social and economic conditions and the decision taken by healthcare workers to migrate for better employment opportunities has been documented in the international literature (Hashish & Ashour, 2020). However, little is known about migration patterns of highly skilled individuals to the United States, particularly those coming from U.S. territories such as Puerto Rico. For instance, the rapid out-migration of Puerto Rican citizens is gaining more attention due to recent socio-political and environmental crises experienced on

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the island (Caban, 2021). Over 130,000 islanders migrated to the mainland United States only in 2017, leading to a population drop of 11% Puerto Ricans in nearly a decade (Abel & Deitz, 2014). Currently, there is a gap in knowledge about the “brain drain” phenomenon among mental health professionals worldwide, a group recognized for its critical role in recent global efforts (e.g., the COVID-19 pandemic), and by the World Health Organization as essential healthcare workers (Weir, 2020). In Puerto Rico, it is particularly important to have a strong mental health workforce in the context of recurrent natural disasters impacting this vulnerable population.

Mental Health Professionals in the United States

Mental health professionals are an essential healthcare workforce in the mainland U.S. and in Puerto Rico. Through rigorous training and graduate education, these professionals are equipped to identify and engage in evidence-based practices that are effective in decreasing adverse outcomes among vulnerable populations (Giraldo-Santiago et al., 2023; Lamers et al., 2012). These professionals consist of psychologists, social workers, and counselors, among others, who provide services in a wide range of settings, including schools, communities, and mental health and medical settings. The growing prevalence of mental health needs among individuals across their lifespan has created severe shortages in services that are attracting mental health professionals to pursue these careers in the United States.

In 2022, President Biden announced the *American Rescue Plan*, a national mental health strategy to strengthen the capacity of systems to address the severe shortages of mental health professionals in the United States (The White House 2022). The plan highlights the need to dramatically expand the supply, diversity, and cultural competency of the mental health workforce—from psychiatrists to psychologists, peers to paraprofessionals—and increase opportunities and incentives for them to practice in areas with the highest need. The *American Rescue Plan* has prioritized the promotion of frontline workers’ well-being, with around \$103 million designated to address burnout and strengthen resiliency. While U.S. Congress efforts to improve the well-being of the mental health workforce are indispensable, it is often unclear how these are translated in Puerto Rico, given their socio-political and geographic location. For example, government officials and the media have documented the challenges faced by Puerto Ricans in understanding and accessing federal programs meant to aid recovery (U.S. Government Accountability Office, 2022).

Puerto Rico’s Sociopolitical and Economic Status

Different from other Latin American origin immigrants, Puerto Ricans can travel freely back and forth between the Island and the mainland. Puerto Ricans may be perceived as privileged given their U.S. citizenship status from birth. However, they lack representation in the U.S. Congress, voting powers, and some social safety net programs compared to U.S. citizens living in the mainland United States. Puerto Ricans’ territorial and colonial position has worsened by recently implemented austerity measures and policies resulting in cuts to healthcare and social services (Bonilla, 2022; Makoff & Setser, 2017). The impact of such policies has led to social and economic inequalities experienced by Puerto Ricans. A measure of income inequality, where 0 reflects perfect equality and 1 reflects maximal inequality, suggests that Puerto Rico has the highest Gini index (0.542) in the United States (Guzman, 2019). Moreover, poverty rates (43.1%) are higher than the U.S. national rate (13.1%), and more than double that of Mississippi (19.7%), the poorest U.S. state (Glassman, 2019).

Mental Health Needs and Services in Puerto Rico

Mental health professionals are indispensable in Puerto Rico. For example, the high prevalence of mental health needs experienced by Puerto Ricans warrants the maintenance of a workforce that can identify and address such needs (Sánchez-Cardona et al., 2022). However, like the U.S. mainland, there is a shortage of mental health providers in Puerto Rico, where 63 of the 78 geographic areas or municipalities were identified as underserved (Rural Health Information Hub, 2023). Moreover, a recent qualitative study in Puerto Rico suggested that mental health professionals perceive the government as focusing on other issues and neglecting mental health as a public health matter (Negroni et al., 2019). The study found that mental health providers had a sense of frustration about various work-related issues. For instance, providers signaled issues regarding the lack of support from supervisors and organizations to provide mental health services, and training on specialized services for specific populations such as children is limited, among other work constraints. Mental health professionals in Latin America also face limited societal recognition and frequent misconceptions about the essential work of these professionals, resulting in poor income and precarious working conditions.

Socioeconomic and Work-Related Conditions of Healthcare Professionals in Puerto Rico

A study conducted in 2020 with healthcare workers in Puerto Rico during the COVID-19 pandemic found that the overall burnout level of the clinical personnel was moderate

(Cortina-Rodríguez & Afanador, 2020). Moreover, 12.1% of physicians had severe burnout levels compared to a 13.1% of nurses. Additionally, the study found that 92.4% of physicians and 100% of nurses had moderate to severe burnout. Since the beginning of the COVID-19 pandemic, over 90% of healthcare professionals in Puerto Rico have been working with moderate to severe burnout, with nurses being the most affected (Cortina-Rodríguez & Afanador, 2020). A recent qualitative study with nurses in Puerto Rico examined the personal and work-related stressors of healthcare workers and the organizational support they received during the pandemic. Nurses pointed out that the staff shortage contributed to nurses assuming greater workloads, feeling overwhelmed and overworked, and resignations (Alvarez et al., 2023).

Little is known about emotional exhaustion among mental health professionals in Puerto Rico, and whether this contributes to their intent to migrate. A recent study conducted with an interdisciplinary sample of mental health professionals in Puerto Rico found that almost half (44%) reported having more than two jobs and incomes below \$30,000 a year (40.6%; Giraldo-Santiago et al., 2023). However, the study of how socioeconomic factors are associated with the migration of mental health professionals (e.g., psychologists, counselors, and social workers) remains unclear.

Drivers for Migration Among Health Workers

International literature has found that healthcare professionals (general health practitioners and nurses) reporting intention to emigrate were mostly male, young, not married, and more educated (Ferreira et al., 2020). Moreover, the study found that those perceiving their income as insufficient for their needs and showed higher levels of burnout at work and professional dissatisfaction were more likely to have thoughts of migration. Consistent with other literature, migration appears to be more common among males and those 31 years old or younger, and single. Among the reasons for migration were having a fair level of workload and job promotion, suggesting that the migration of health professionals is situation dependent, mediated by basic socio-demographic variables and work-related conditions (Lowe & Chen, 2016).

Moreover, perceived organizational support (POS) has been found to significantly influence employees' intention to stay in an organization (Nargotra & Sarangal, 2021). Still, the link between work-related conditions (e.g., limited incomes and perceived organizational support) and thoughts of migration are not clear in the literature. Efforts to understand the prevalence and factors associated with the migration of mental health professionals in Puerto Rico are crucial to know given the ever-growing mental health needs in the Island, and increased migration patterns to the United States

will mean that the mental health needs gaps will possibly widen.

The Current Study

This study sought to examine whether socioeconomic variables (i.e., age, gender, marital status, highest level of education, income satisfaction) and work-related factors (i.e., emotional exhaustion and perceived organizational support) lead to thoughts of wanting to migrate to the U.S. mainland among mental health professionals in Puerto Rico?

Methods

Sample

The target population of this study consisted of a convenience sample of practicing licensed social workers, psychologists, and professional counselors in Puerto Rico. Participants were recruited through email correspondence via three professional associations in Puerto Rico: the College of Social Work Professionals in Puerto Rico (CPTSPR), the Professional Counselors Association (CPCR), and the Psychology Association of Puerto Rico (APPR).

Procedure

Data were collected between October and December 2021. Participants had to be residing in Puerto Rico, identify as Latino (Mexican, Puerto Rican, Cuban, Central, or South American), be licensed to practice social work, counseling, or psychology, and be employed part-time or full-time in a public or private agency in Puerto Rico. An online survey through Qualtrics was used to collect the data. The survey took approximately 10 to 15 min to complete and was self-administered. The survey was developed in Spanish and was available in English. Participants were compensated with \$ 5 for their participation. To increase response rates, the research team sent follow-up emails, and engaged in several recruitment strategies, such as in-person data collection at a social work conference in Puerto Rico.

Measures

The variables utilized included individual and organizational domains that were important to identify “push” and mitigating factors related to the migration of mental health professionals.

Thoughts of Migrating to the United States

The outcome (dependent) variable of the study consisted of a single-item question, “*During the past 12 months, have you thought of moving to the U.S. mainland?*” Responses to this question were analyzed categorically and coded as 1 for respondents who thought of migrating in the last year (yes) and 0 for those who did not think of migrating (no).

Socio-Demographic Factors

Demographic variables were considered determinants and mitigating factors related to the migration of mental health professionals. These variables included age as a continuous variable. Gender was converted in a dichotomous variable with male (0) and female (1) as only two participants identified with the following options: non-binary, gender not listed, and prefer not to answer. Also, marital status was re-coded from five categories to a binary variable that grouped married and cohabitating as 1 vs. single, divorced, and widowed individuals as 0. As this was a sample of licensed mental health professionals with graduate studies, a variable examined whether the participant had a master’s, a doctoral degree, or something else. These responses were dummy-coded into three variables to compare each group. Another relevant variable gathered information about practitioners’ professional backgrounds and whether they identified as social workers, counselors, or psychologists. This variable was also dummy-coded before being entered into the regression model. A measure of economic hardship was used to describe the needs of the sample with one question: “How would you describe your financial circumstances in general?” Response options ranged from (1) I do not have to worry about money, (2) I have extra money after paying bills, (3) I am breaking even, (4) I am barely making it, and (5) I cannot make ends meet.

Organizational Factors

Satisfaction with current salary was measured with a five-point Likert scale, with the following question: “How satisfied are you with the salary you currently receive for the work you do?” Responses consisted of 1 = Very Dissatisfied, 2 = Not very Satisfied, 3 = Moderately Satisfied, 4 = Satisfied, and 5 = Very Satisfied. A higher score suggested more satisfaction with salary.

Perceived Organizational Support (POS)

The mean of the POS-8 items scale (Eisenberger et al., 1997) included items such as (1) The organization values my contribution to its well-being; (2) The organization fails to appreciate any extra effort from me; (3) The organization

would ignore any complaint from me; (4) The organization really cares about my well-being; and (5) The organization cares about my general satisfaction at work, among others. This measure of POS has been found to be a strong measure of employees’ perceived support from their employer. Likert scale ranged from strongly disagree (1) to strongly agree (5). Higher scores represented higher POS. Negatively worded items were reversed coded. The Cronbach Alpha reliability coefficient of this scale was 0.89.

Emotional Exhaustion

Emotional exhaustion was measured by the sum of a 3-item subscale of the Maslach Burnout Inventory (MBI-9 scale) (Maslach & Jackson, 1981). This subscale included the following items: (1) Working with people all day is really a strain for me; (2) I feel I am positively influencing other people’s lives through my work; and (3) I feel fatigued when I get up in the morning and have to face another day on the job. Responses ranged from 1 = Never to 6 = Every day, and higher scores represented more emotional exhaustion in the workplace experienced at the time of survey completion. The emotional exhaustion subscale had adequate reliability of 0.87.

Data Analysis

IBM SPSS version 29 was used to perform data analysis. Univariate descriptive statistics were run to describe and summarize the data. The range of values and measures of central tendency were explored to know how data was distributed. Variable distribution in all scales was checked. Model assumptions, such as non-linearity and excessive collinearity, were examined (Menard, 2002). First, frequency distributions of all variables were computed, and second, selected socio-economic and work-related variables (age, gender, marital status, income satisfaction, highest level of education, and perceived organizational support, thoughts of migration, among others) were cross-tabulated with each other. Bivariate descriptive statistics were used to analyze, compare variables, and determine whether they were significantly correlated, and mean differences were explored. Cohen’s standard was used to evaluate the strength of the relationships, where coefficients between 0.10 and 0.29 represented a small effect size, coefficients between 0.30 and 0.49 represented a moderate effect size, and coefficients above 0.50 indicated a large effect size (Gignac & Szodarai, 2016). The result of the correlations was examined using the Holm correction to adjust for multiple comparisons based on an alpha value of 0.05. This study sought to explore the socioeconomic and work-related factors related to the migration of mental health professionals in Puerto Rico. Thus, a binary logistic regression was performed on

seven independent variables (i.e., age, gender, marital status, occupation, highest degree achieved, income satisfaction, emotional exhaustion, and perceived organizational support) to assess their impact on the likelihood that subjects would fall in the binary category (yes/no) of having thoughts about migrating, the associated outcome variable. Univariable and multivariable logistic regression modeling were utilized to estimate unadjusted and adjusted odds ratios (ORs). This was conducted to account for the influence of each independent variable when entered into a model altogether.

Results

Descriptives

The sample consisted of 237 practicing licensed social workers, psychologists, and professional counselors living and working in Puerto Rico. Almost half of the participants were social workers ($n = 45.6\%$), followed by psychologists (29.1%) and professional counselors (25.3%). Participants were mostly female (83.4%) and the mean age was 42.92 ($SD = 11.705$). Almost half of participants reported being married (46%). Close to three quarters had a master's degree (63.4%), followed by a doctoral degree (26.4%). Annual income was significantly diverse: 7.7% earned less than 20 K, 32.9% between 20 and 30K, 26.1% between 31 and 40K, and 14.1% between 41 and 50K. See Table 1 for more details.

Salaries and Number of Jobs by Occupation

Table 1 presents salaries reported by the total sample. The most frequent salary category ranged between \$21,000–30,000 a year, with (32.9%) of professionals belonging to this group. Additional analyses (not reported in tables) suggested significant differences in salary across social workers, psychologists, and professional counselors $\chi^2(235) = 76.52$, $p = 0.000$. Social workers and counselors were more likely to earn between \$20–30K a year. Meanwhile, psychologists had greater salaries, with the largest group earning between \$ 41–50K (29.3%), followed by another group earning between \$ 51–60K (20.3%), and 15.9% earning more than \$60K. These differences in salaries can be explained by significant differences in the number of jobs reported by these professionals ($\chi^2(235) = 21.62$, $p = 0.000$). Psychologists were the largest group with two or more jobs (17.9%), followed by professional counselors (13.2%) and social workers (12.8%).

Table 1 Demographic characteristics of the sample

Characteristics	Values (Percentages)
Age	
Mean (SD)	42.92 (11.705)
Gender	
Female	196 (83.4)
Male	39 (16.6)
Marital Status	
Single	67 (28.5)
Married	109 (46.4)
Divorced	27 (11.5)
Cohabiting	24 (10.2)
Widow	8 (3.4)
Highest education	
Master's degree	149 (63.4)
Doctorate degree	62 (26.4)
Something else	24 (10.2)
Income	
Less than 20K	18 (7.7)
\$21,000–30,000	77 (32.9)
\$31,000–40,000	61 (26.1)
\$41,000–50,000	33 (14.1)
\$51,000–60,000	19 (8.1)
More than 61K	17 (7.3)
Prefer not to answer	10 (4.3)

Income Satisfaction

The income satisfaction mean for all respondents was 2.48 of 5.00 ($SD = 1.23$). An analysis of variance (ANOVA) revealed significant differences in income satisfaction by those who had thoughts of migration ($M = 2.80$, $SD = 1.24$) and those who did not ($M = 3.27$, $SD = 1.42$) $F(1,22) = 6.96$, $p = 0.009$. In other words, the income satisfaction of those who had thoughts of migrating was lower.

Emotional Exhaustion

The sample had a mean of emotional exhaustion of 2.94 ($SD = 1.35$), and this was higher among those who had thoughts of migrating in the last year but not significantly different. Significant differences were found across occupations, with social workers having the highest mean of emotional exhaustion ($M = 3.22$), followed by professional counselors ($M = 2.75$) and psychologists ($M = 2.66$) ($F(15, 220) = [F-1.770]$, $p = [0.040]$).

Correlations Between Key Variables

Table 2 presents correlations between key variables. The correlation coefficient (r) between age and greater perceived

organizational support was $r=0.182$, $p=0.06$, $n=237$, indicating that older age was correlated with higher POS scores. Meanwhile, the correlation between emotional exhaustion and greater POS was $r=-0.525$, $p<0.001$, $n=237$, suggesting the presence of less work exhaustion in more supportive organizations. Similarly, thoughts of migrating to the United States were associated with age, emotional exhaustion, perceived organizational support, and income satisfaction, but not with gender, marital status, and highest degree achieved.

Demographic Characteristics by Thoughts of Migration to the United States

A total of 115 (48.9%) participants reported having thoughts of migration in the last year. Significant differences ($F=16.53$; $p<0.000$) were found between having migration thoughts and age and income satisfaction, while no significant differences were observed by occupation, marital status, or emotional exhaustion. Table 3 presents differences in the sample of professionals with and without thoughts of migration across each of the key variables. The mean age was less among those in the group

Table 2 Correlations of key variables

Variables	1	2	3	4	5	6	7	8
1. Age		0.06	0.01	0.07	-0.09	0.18*	-0.07	0.13*
2. Gender			0.00	0.04	0.01	-0.09	-0.11	-0.07
3. Marital Status				-0.15	0.04	-0.10	-0.17*	-0.10
4. Degree					-0.02	0.15*	0.14*	0.11
5. Emotional Exhaustion						-0.53*	-0.25*	-0.21*
6. Perceived Organizational Support							0.36*	0.16*
7. Income Satisfaction								.180*
8. Thoughts of migration								

Note *Significant correlations

Table 3 Demographic characteristics by thoughts of migration to the United States

Variable	Thoughts of migrating in the last year YES	No thoughts of migrating in the last year NO	p = value
Age	41.34 (9.521)	44.25 (13.229)	$p<.001$
Marital Status			$p<.092$
Single, divorced, widow	50 (54.9%)	41 (45.1%)	
Married, cohabitating	65 (45.1%)	79 (54.9%)	
Occupation			$p=.880$
Social Workers	54 (50%)	54 (50%)	
Psychologists	32 (46.4%)	37 (53.6%)	
Professional counselors	29 (50%)	29 (50%)	
Emotional exhaustion	3.2290 (1.26689)	2.6639 (1.37369)	$p=.466$
Income satisfaction			$p<.001$
Very satisfied	2 (14.3%)	12 (85.7%)	
Satisfied	13 (33.3%)	26 (66.7%)	
Moderately Satisfied	25 (42.4%)	34 (57.6%)	
Not very satisfied	33 (57.9%)	24 (42.1%)	
Very dissatisfied	44 (63.6%)	24 (36.4%)	
Economic hardship			$p=.005$
I don't have to worry about money	6 (5.2%)	14 (11.7%)	
I have extra money after paying bills	24 (20.9%)	45 (37.5%)	
I am breaking even	44 (38.3%)	35 (29.2%)	
I am barely making it	29 (25.2%)	21 (17.5%)	
I cannot make ends meet	12 (10.4%)	5 (4.2%)	
Perceived Organizational Support	2.9783 (.81788)	3.2417 (.83164)	$p=.701$

having thoughts of migration. In other words, professionals in the group having thoughts of migration were more likely to be younger ($M = 41.34$) when compared to their counterparts ($M = 44.25$). Emotional exhaustion was less present in the group with no thoughts of migration, but it was not statistically different from those with thoughts of migration. Similarly, perceived organizational support was greater among those belonging to the group with no thoughts of migration in the past year, but the difference was not statistically significant.

Regression Model

The regression model was statistically significant, $\chi^2(1) = 32.68$, $p = 0.692$. The Cox-Snell R-squared for the model was 0.13, and the Nagelkerke R-squared for the same model was 0.18. Thoughts of migration to the United States were associated with age, income satisfaction, and emotional exhaustion. For instance, for one SD unit difference in age, the odds of migrating were lower by 0.709 (OR per year = 0.709, $p = 0.026$). In other words, for one unit increase in age, the likelihood of having thoughts of migration decreases by 1.41. Moreover, greater emotional exhaustion increased the odds of having thoughts of migration compared to those who were not emotionally exhausted (OR per unit = 1.328, $p = 0.024$).

Similarly, those with greater satisfaction with income were 1.59 times more likely not to have thoughts of migration (OR per Likert unit = 0.627, $p < 0.001$) compared to those with less satisfaction. At the univariate level, findings suggested that greater perceived organizational support decreased the odds of having thoughts to migrate (OR per unit = 0.697, $p = 0.016$). In other words, for those who perceived lower organizational support, the odds of wanting to migrate were 1.47 greater than those in supportive organizations. Gender, marital status, type of

degree achieved, and occupation were not factors related to migration thoughts (Table 4).

Discussion

This study examined socioeconomic and work-related factors associated with thoughts of migration to the U.S. mainland among mental health professionals in Puerto Rico. Findings suggest that younger professionals, those with higher income dissatisfaction, greater emotional exhaustion, and those who perceived limited organizational support were more likely to think of migrating to the U.S. in the prior year. Results from this study are consistent with the existing literature related to socioeconomic factors (Hajian et al., 2020), emotional exhaustion (Györfy et al. 2018), and work-related stressors (Almansour et al., 2023) influencing the presence of migration thoughts among healthcare professionals. To our knowledge, this is the first study to examine the push and pull factors of migration among mental health professionals. This study builds on the limited literature on the importance of preventing health workforce shortages that contribute to burnout, excessive workload, dissatisfaction, and consequently, to migration (Domagała and Dubas-Jakóbczyk, 2019).

Consistent with the international literature focused on healthcare professionals (Ferreira et al., 2020), younger mental health professionals were more likely to have thoughts of migration than their older colleagues, a finding that exacerbates concerns about long-term economic development and the implications of an older workforce addressing the growing mental health and health needs of the Puerto Rican population (Matos-Moreno et al., 2022). The rapidly aging population of Puerto Rico, which includes healthcare professionals, warrants generating strategies and interventions to address their socioeconomic and professional needs. This is important, as older mental health providers have been found

Table 4 Unadjusted and Adjusted Odds Ratios of Univariate and Multivariate Regression Models Associated with Thoughts of Migration

	Unadjusted Odds Ratio (OR)	p-value	Adjusted Odds Ratio	p-value
Age	.776	.059	.709	.026
Gender (male vs. female)	.383	.281	1.413	.383
Marital Status (single vs. other)	1.482	.144	.805	.455
Degree (Master's)	1.408	.439	1.140	.793
Degree (Doctorate)	.697	.458	.605	.426
Emotional Exhaustion	1.379	.002	1.328	.024
Perceived Organizational Support	1.47	.016	2.44	.409
Income Satisfaction	.616	.001	.627	.001
Occupation (Ref: Social Worker)	1.000	1.000	.424	.424
Occupation (Ref: Psychologist)	.865	.684	.916	.916

Note Z-scores for age were calculated for a better interpretation of the odds ratio

to have less interest in the appeal of empirically supported interventions and the likelihood of adopting empirically supported interventions if they were required to do so (Giraldo-Santiago et al., 2023). It is possible that mental health professionals who remain on the island are not only growing older but also absorbing the caseload of those leaving for the US mainland. The potential continued migration of young mental health professionals will leave a gap in services to the most marginalized, including older adults, as the percentage of this population continues to rise in Puerto Rico.

Migration thoughts were strongly predicted by dissatisfaction with incomes. In this study, more than half of the sample (59%) had salaries that ranged between 21,000 and 40,000 a year, a compensation that is significantly less than the average salary of mental health workers U.S. mainland (above 50K; U.S. Bureau of Labor Statistics, 2021). The Substance Abuse and Mental Health Service Administration (2022) have recently recommended organizations to implement incentives and compensation strategies to improve satisfaction and employee's well-being. Moreover, raising mental health worker's wages, particularly among those from historically under-represented communities, should be prioritized as the cost of living continues to increase and incomes are not (Rivera Sanchez, 2021). Another plausible approach to decelerate the "brain drain" in this field is to provide incentives (such as tax credits or exemptions) for the relocation of mental health professionals back to the Island. For example, the federal government incentivize the relocation of mental health professionals in tribal and rural parts of the U.S. A similar approach could be viable to avoid a catastrophic collapse of the mental health mainstream system in Puerto Rico.

Emotional exhaustion in the workplace and limited perceived organizational support were also at the forefront of the phenomenon of out- of- the Island migration. This lack of perceived support in the workplace (Negroni et al., 2019) and emotional exhaustion could significantly impact the quality of interventions and service delivery in the long term. As recommended by the Substance Abuse and Mental Health Service Administration (2022), there is a need to address excessive workloads, insufficient compensations, among other factors related to the organizational culture and climate. Examples of burnout experienced by healthcare workers have been documented, particularly becoming more acute during the COVID-19 pandemic (Cortina-Rodríguez et al., 2020) around the same time this study was conducted. In this study, social workers emerged as more vulnerable than professional counselors and psychologists to experiencing emotional exhaustion. Future research and interventions should target health services and organizational settings to address working conditions, needs, and perceived attitudes related to the workplace across all disciplines.

Different from what we hypothesized and prior literature (Ferreira et al., 2020), gender, marital status, and higher degree accomplished (doctorate vs. master's degree) did not have an influence on thoughts of migration among Puerto Rican mental health professionals. It can be inferred that individual decisions, such as attempts to obtain higher educational degrees, may not serve as protective factors against thoughts of migration from Puerto Rico, as found in other studies. The present study depicts a picture of healthcare professionals who are satisfied with their careers but struggle with unfavorable job conditions and organizations, which could become barriers to the provision of best services. Findings suggest the need to increase organizational strategies that decrease emotional exhaustion, dissatisfaction with salaries, and lack of perceived support from employers in Puerto Rico. The provision of opportunities and resources to increase personal accomplishment has been linked to employee's satisfaction (Siefert et al., 1991). Future research could explore whether mental health professional that have already relocated to the United States mainland would consider returning to PR if the working conditions are comparable to what they are offered in the mainland.

Puerto Ricans on the island have historically experienced marginalization and disinvestment due to histories of colonialism, which has negatively impacted the economy (Joseph et al., 2020). Policies geared towards improving the training, retainment, and compensation of island-dwelling practitioners should be prioritized. For example, a Behavioral Health Workforce Act in Nebraska designated funds to support a behavioral health workforce center, which resulted in an increase in the mental health workforce and increased additional trainings subsidized by the government (Nebraska Legislature, 2022). This approach could allow Puerto Ricans workers to expand their skillset, create a wider network of mental health professionals, and improve the compensation structures of highly skilled clinicians in the island. Efforts to retain bilingual and bicultural mental health professionals in the island is crucial for the health of the larger Puerto Rican community, as widening the gap in mental healthcare due to the mass exodus will only exacerbate the existing health and economic disparities for Puerto Ricans.

Mental health professionals living and working in Puerto Rico are facing tremendous needs and challenges that threaten the continuation of accessible mental health services (Giraldo-Santiago et al., 2023). The growing migration out of Puerto Rico and the burnout experienced could be attributed to multiple factors, such as the government's inadequate response to the socioeconomic and health-related needs left by Hurricane Maria in 2017 (Glassman, 2019). Although many Puerto Ricans have resisted and manifested against this systematic oppression

and lack of resources (Alvarez-Hernandez, 2020), it is evident that thoughts of migration remain a means to escape the systematic oppression and disadvantaged experienced in organizational settings.

Implications and Recommendations

The findings of this study have various implications for the retention of mental health professionals in Puerto Rico and the potential reception of these professions in the U.S. mainland post-migration that warrant some recommendations. For the retention of mental health professionals in Puerto Rico, our findings underscore the importance of developing policies focused on retaining young mental health professionals. One potential strategy is for the Puerto Rican government to establish policies that incentivize mental health practice in Puerto Rico by increasing salaries across these professionals. At the organizational level, allocating resources to address emotional exhaustion and the lack of perceived organizational support is indispensable to decrease thoughts of migration. Some of these resources may include capping the number of cases per provider, offering hybrid and telework options, and providing competitive salaries.

When considering the potential reception of mental health professionals migrating from Puerto Rico to the U.S. mainland, it would be necessary to ensure access to professional licensing in the state of arrival and adequate job opportunities as prior literature suggests the presence of many structural barriers for Puerto Rican migrants who want to practice in the U.S. mainland (Ninow et al., 2022). It is also critical for organizations hiring professional migrants to explore their needs and experiences and identify ways for support. For example, it would be worth exploring the impact of socio-economic, interpersonal, and structural factors (e.g., the role of language, social capital, and wages) on their practice with new populations (Bernabe-Aguilera 2005). Another area of concern for the practice and education of social work and mental health disciplines such as psychology and counseling, is to ensure that access to licensing and adequate job opportunities in the US as prior literature suggests the presence of many structural barriers for Puerto Rican migrants who want to practice in the United States are failing to receive language accommodations or resources to facilitate this process (Ninow et al., 2022).

Limitations

The present study had certain limitations. This cross-sectional study did not allow for more complex causal inference analysis of the outcome variable. A longitudinal study would be preferable to examine the complex relationship between the independent and dependent variables. However, we believe that this study serves as a foundation for future

studies examining this brain drain phenomenon using more rigorous analyses and methodologies. Qualitative interviews could be ideal to gain a better understanding of the lived experiences and the factors contributing to the migration of mental health professionals in Puerto Rico. Another limitation of this study is the inability to generalize the findings, because the sample selection was based on convenience. As observed, the sample was not equally stratified, with an over-representation of social workers when compared to psychologists and professional counselors.

Conclusion

The migration of healthcare workers is a major aspect of globalization and a phenomenon creating circumstances in which some countries gain, and others lose opportunities for social and economic development (Docquier & Rapoport, 2011). Our study added to the current literature on the socio-economic and professional factors contributing to thoughts of migration by many mental health professionals in Puerto Rico struggling with precarious living and working conditions and limited earnings. We provided policy and organizational-level recommendations to retain mental health professionals in Puerto Rico and potentially receive them in the U.S. mainland. Such efforts are crucial for the health of the larger Puerto Rican community, as widening the gap in mental healthcare due to the high patterns of migration will only exacerbate the existing health and economic disparities for Puerto Ricans in Puerto Rico and the U.S. mainland.

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